

CONTAINERIZATION AND DOCKER BASICS

❖ Difference Between Monolithic and Microservices

Feature	Monolithic Architecture	Microservices Architecture
Definition	Single, unified application where all components are tightly coupled.	Application is divided into smaller, independent services.
Structure	One large codebase.	Multiple small, modular services.
Development	Difficult to develop and maintain as it grows.	Easier to manage, each service can be developed independently.
Deployment	Deployed as a single unit.	Each service can be deployed independently.
Testing	End-to-end testing is easier.	Requires extensive integration testing.

❖ Difference Between Traditional, Virtualization, and Containerization Deployment

Feature	Traditional Deployment	Virtualization	Containerization
Concept	Apps run on physical servers	Multiple VMs on single host	Multiple containers share OS kernel
Resource Utilization	Low	Better than traditional	Very efficient
Isolation	Limited	Full OS-level isolation	Process-level isolation
Boot Time	Minutes to hours	Minutes	Seconds
Portability	Difficult	Limited by hypervisor	Highly portable
Overhead	High	Medium	Low

❖ Image:

A static file with executable code that creates containers. It includes application code, runtime, system tools, libraries, etc.

Key Benefits:

1. Consistency across environments
2. Faster deployments
3. Scalability Easier rollback and
4. Upgrades

❖ Introduction to Docker

Docker is an *open-source* platform for *developing*, *shipping*, and *running* applications inside containers.

✓ Key Components:

- **Docker Engine:** The runtime that runs and manages containers.
- **Docker Image:** The blueprint used to create containers.
- **Docker Container:** The running instance of an image
- **Dockerfile:** A script with instructions to build Docker images.
- **Docker Hub:** Cloud-based registry to store Docker images.

❖ Introduction to Containerization, Containers, and Images

❖ Containerization:

A lightweight form of virtualization that packages an application and its dependencies into a container, enabling consistent and efficient deployment.

❖ Container:

A running instance of a container image. Isolated from the host and other containers. Lightweight and fast.

Difference Between Docker CE and Docker EE

Feature	Docker CE (Community Edition)	Docker EE (Enterprise Edition)
License	Free & open-source	Paid, commercial
Support	Community-based	Official enterprise-grade support
Features	Core Docker features	Additional security, management, and orchestration tools
Use Case	Development, learning	Production, enterprise use

❖ Install Docker Engine

1. Enter url on google

<https://docs.docker.com/engine/install/ubuntu/>

2. Copy & past terminal

Add Docker's official GPG key:

```
sudo apt-get update
```

```
sudo apt-get install ca-certificates curl
```

```
sudo install -m 0755 -d /etc/apt/keyrings
```

```
sudo curl -fsSL
```

```
https://download.docker.com/linux/ubuntu/gpg -o
```

```
/etc/apt/keyrings/docker.asc
```

```
sudo chmod a+r /etc/apt/keyrings/docker.asc
```

Add the repository to Apt sources:

```
echo \
```

```
"deb [arch=$(dpkg --print-architecture) signed-  
by=/etc/apt/keyrings/docker.asc]
```

```
https://download.docker.com/linux/ubuntu \
```

```
$(. /etc/os-release && echo
```

```
"${UBUNTU_CODENAME:-$VERSION_CODENAME}")
```

```
stable" | \
```

```
sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
```

```
sudo apt-get update
```

```
sudo apt-get install docker-ce docker-ce-cli
```

```
containerd.io docker-buildx-plugin docker-compose-
```

```
plugin
```

▪ Verify Installation:

```
✓ docker --version
```

```
✓ sudo systemctl status docker
```

[GENERAL INTERVIEW QUESTIONS]

Q1. Difference between Monolithic and Microservices

1. What is a Monolithic application?

- A *Monolithic* application is a single-tiered software application in which all components are combined into one program from a single platform

2. What is a Microservices-based application?

- A *Microservices* application is an architectural style where the application is composed of small, independent services that communicate over APIs.

Q2. Key difference between Monolithic and microservices?

- **Monolithic** is *tightly* coupled, deployed as a *single* unit.
- **Microservices** are *loosely* coupled and deployed *independently*

Q3. Difference between Traditional, Virtualization, and Containerization Deployment

❖ Traditional deployment:-

- Each application runs on a dedicated physical server, often leading to underutilization of resources.

❖ Virtualization:-

- It allows multiple virtual machines (VMs) to run on a single physical server using a hypervisor.

❖ Containerization:

- A lightweight form of virtualization that packages an application and its dependencies into a container, enabling consistent and efficient deployment.

❖ Main differences among the three?

- **Traditional** uses full hardware,
- **virtualization** uses VMs with OS,
- **containerization** shares the OS and uses isolated processes

❖ Difference between Docker CE and Docker EE s

Q1: What is Docker CE?

- Docker Community Edition (CE) is a free version for developers and small teams.

Q2: What is Docker EE?

- Docker Enterprise Edition (EE) is a commercial version with advanced features, support, and security.

Q3: Key difference between Docker CE and Docker EE?

- CE is free and open-source. EE is paid with enterprise support and advanced capabilities